

Exploring the Impact of Domestic Robotics on Preventing Lonely Deaths among Japan's Hikikomori

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ABSTRACT

This research focuses on the potential of domestic robotics to address the declining labor force in elderly care and the challenges faced by Hikikomori individuals in Japan. Hikikomori are people who socially withdraw, often for extended periods, and struggle to reintegrate into society. The decline in the birth rate and the increasing prevalence of social withdrawal pose significant challenges. Traditional methods of support face resistance, making it crucial to explore alternative approaches, such as the use of domestic robotics. This research examines the current approaches and potentially relevant results from similar research areas.

CCS CONCEPTS

• **Applied computing** → **Consumer health**; • **Computing methodologies** → *Robotic planning*; • **Human-centered computing** → *Interactive systems and tools*.

KEYWORDS

hikikomori; social withdrawal; lonely death; robotics; care; HRI

1 INTRODUCTION

Japan is currently confronted with four major and continually growing challenges: it is one of the world's most aged societies, the birth rate is decreasing while one-person-households are increasing, the labor force is shrinking, and the phenomenon of social withdrawal is increasing. Social withdrawal, which is related to the other three challenges, refers to an estimate of over one million people who have stopped going to school, university or work, retreating to their homes or in extreme cases to their bedrooms, leaving mostly only at night to get essentials from a convenience store. In worst cases, even contact with their family living under the same roof may cease. The Japanese call these people Hikikomori, a term used for both the phenomenon and the individual. Individual cases vary regarding the nature of withdrawal and the reasons behind it, but all have in common the avoidance of social contact.

The Japanese way of life has a set of rigorous guidelines for how a person's academic and job prospects should unfold. Often, young people who are not able to fit into the inflexible system experience a sense of failure [29].¹ It is proposed that the initial decision to withdraw arises from a desire "to stop growing up" [5], as a protest

against the system, from cultural grounds or as a coping strategy to stress [5, 10]. There are non-profit and governmental organizations aiming to help the hikikomori; for example, with online or group therapy sessions or job offerings, such as cleaning and selling donated books online. However, numerous extreme Hikikomori do not want to seek or accept help, which is also rooted in Japanese culture, where it is perceived shameful not to be successful, resulting in not wanting to be a burden [22].

Herein lies the problem of lonely deaths [*kodokushi*]: a death in solitude, which remains undiscovered for a significant amount of time – not rarely up to several months [22]. These solitary deaths are the foundation for Japan's increasingly prevalent, 8050-problem. The first wave of young hikikomori from the 1990s have aged, resulting in a percentage of 1.45% of 40-64 year olds (613,000 people) [2]. The issue is that their parents, who often provide and supply their hikikomori children with food and money, are also aging and entering their 80s – resulting in what is known as the 8050 problem.

This is a time-critical issue, resulting in hikikomori being unable to care for themselves once their parents have passed away. The result of their parents passing away varies from having to interact with the world outside their home for the first time in decades, to not being able to afford food because they do not fulfill the requirements for receiving a pension. The latter can and does result in a lonely death, caused by starvation or extensive lack of nutrition [33].²

Given that this problem is rooted in a decade-long rejection of society, it is very resistant to traditional intervention methods [33]. This paper focuses on the potential impact of a new approach, using domestic robotics, which is an evolving field, in preventing tragic deaths resulting from social isolation.

2 METHODOLOGY

This research followed three main approaches to evaluate the possible impact domestic robotics could have on hikikomori. 1) The most popular hikikomori definitions and categorization attempts were analyzed. 2) The subsequent focus was on research done specifically on or for hikikomori that utilized technology or robotics. 3) We then examined research that wasn't directly related to hikikomori to uncover potential connections – for example, we examined Japan's robotic elderly care.

The objective was to determine how much of Japan's advanced robotics expertise could be applied to hikikomori. Particularly considering the fact that hikikomori are also aging and will require

¹Evelyn Schulz. 2022. Hikikomori: Disappearing quietly. Retrieved July 12, 2023 from <https://www.lmu.de/en/newsroom/news-overview/news/hikikomori-disappearing-quietly.html>



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²NHK. 2022. Dying Out of Sight: Hikikomori in an Aging Japan. Retrieved July 12, 2023 from <https://www3.nhk.or.jp/nhkworld/en/tv/documentary/20220703/4001383>

elderly care, which, we assume, should contain minimal social interactions. Given these approaches, the following search term was created:

*(robotics AND (hikikomori OR (isolation AND loneliness))
OR hikikomori) AND japan*

“Isolation AND loneliness” was used as a substitution for “hikikomori”. Conjugated with “robotics” we aimed to retrieve research relevant to approach two and three. The second ‘hikikomori’ occurrence was added as a disjunction to retrieve results relevant to approach one. Combining all this with “japan” purposes at increasing the result’s precision.

After searching the databases PubMed and ACM, 278 records were identified (Fig.1).

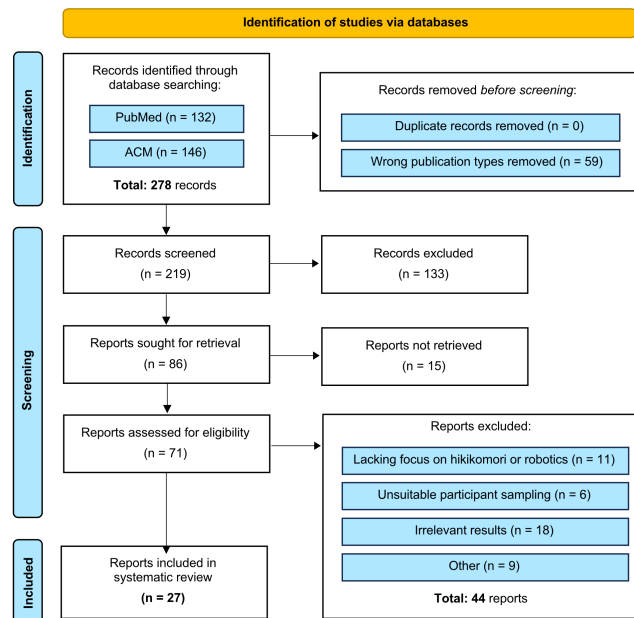


Figure 1: Diagram of the data collection process, conducted according to the PRISMA guidelines

Before screening, 59 records were removed because of wrong publication types, such as proceedings. No duplicates were found automatically. After screening titles and abstracts, 133 records were excluded, and 177 reports continued to get reviewed with their full-texts – 15 reports could not be retrieved. In the assessment for inclusion, the main exclusion criteria were irrelevance, irrelevant results, unsuitable participant sampling and too little focus on hikikomori or relevant robotics. The remaining 27 reports included in this review were exported into a citation and full-text manager which can co-operate with a backlink-note-taking program, enabling simultaneous categorization and data extraction, that could be analyzed visually through a graph system, due to the backlink-functionality.

3 RELATED WORK

3.1 Hikikomori definitions, origins, and effects

The term hikikomori has been spreading and developing since the 1990s, when it was first defined as describing those withdrawing to their one home for at least six months and for whom other psychiatric disorders do not better explain the withdrawal.[10, 26] Since then, there have been many discussions and proposals of new or modified definitions. Although a consent definition is still to be found, there have been two popular, mostly accepted guidelines.

The first, from the Japanese Ministry of Health, Labour, and Welfare (MHLW), published 2010, contains similar criteria and described it as a “result of various factors”, including being house-bound and withdrawing from social participation outside the home for a period of more than six months. However, it contains more details regarding psychiatric disorders, stating that it is “considered to be a non-psychotic phenomenon that is distinguishable from [...] schizophrenia”, while not excluding the possibility of a relation [10, 26]. A more current proposal of three diagnostic criteria, from 2019, also requires a continuous social isolation in one’s home for at least six months, however, instead of including psychiatric disorders, the third criteria is a “[s]ignificant functional impairment or distress associated with the social isolation”. These criteria are followed by nine specifiers, that are not mandatory, but useful for characterization, for example: “The individual endorses feeling lonely. The presence of loneliness tends to be more common as the length of hikikomori increases” [10].

Much research has been conducted regarding the classification of hikikomori as psychiatric disorders or as a social issue. One approach is investigating the reasons for the initial withdraw, which has detected multiple factors. Possible withdraw reasoning are coping or avoidance strategies, related to insecure attachments, resulting in denial of their need for connectedness or distrust towards others. Within possibly broken or nuclear families with little support, parents might have had problems teaching empathy and establishing trusting relationships with others. On the other hand, families could tend to “cultivate a parasitical tendency” within their children, building upon Japan’s culturally accepted “overdependence” [amae] [10, 18]. Thus suggesting “coping behaviors to satisfy one’s desire for dependence” [14].

Besides these rather development-based factors, the most common reasons for starting social withdraw are difficulties getting used to their workplace or even finding employment, quitting of work, difficulties with personal relationships or illness [2].

The fact that hikikomori has various staring points and stepping stones, might be an explanation for the difficulty of classification; also regarding psychiatric disorders. However, the literature shows a recommendation towards intervention. Many Hikikomori, especially towards the beginning, feel a relief by escaping their painful realities [11]. They enjoy their peace while watching television, playing video games or by reading books and manga [18].

Nonetheless, as the solitariness extends, most hikikomori begin feeling distress and loneliness [11]. This is the beginning of a downwards spiral – the longer hikikomori stay withdrawn, the more unlikely it is they seek treatment, resulting in a longer withdraw, which again results in a worsening physical wellbeing and them having to face mayor psychological clinical problems [15, 18]. This

is concerning regarding the fact that, of all hikikomori aged 40 to 64, which make up for 1.45% of Japan's population, an estimate of 613,000 people, around 50% have been in a state of hikikomori for seven or more years. Furthermore, there is a tendency for prolonged hikikomori [2].

3.2 Hikikomori, robotics and technology

Since the internet, that we know and use, is equally old as the hikikomori phenomenon and seeing that hikikomori spend most of their time using it, there is a good amount of research investigating their relation to each other. The broad research in this area focuses on the extent of their correlative or causal connection, as well as on the direction of influence – a type of egg and chicken dilemma. If, for example, someone becomes too dependent on the internet, missing out on opportunities to get outside, he may find himself in a social withdrawal state. Conversely, prolonged hikikomori may find themselves using the internet as a “self-help tool to alleviate loneliness” [12, 35].

Pokémon GO, a game that quickly gained global popularity, is being accounted relatively often in reports and research papers for having a positive impact on hikikomori [13, 25]. The AR game is bridging the virtual world of the internet with real-life physical locations. There are reports of hikikomori able to head outside to public parks, driven by self-motivation, to play Pokémon GO, even when previously barely able to go out [13]. However, it is also noted that hikikomori is too psychologically rooted to be solved by solely going outside.

One research has specifically focused on using a humanoid communication robot (CommU) for psychiatric evaluation on hikikomori, by using a teleoperated robot [35]. For each participant, two interviews were conducted – the first by the robot controlled by a psychiatrist and the second by an experienced clinician, who followed the same structure as the robot. Thereafter, the participant filled out a self-rated survey about evaluating their experience. It is important to note that this was a pilot-trial, so only 2 participants took part.

Yet, the results show that continuing in this direction might be helpful. The evaluation of hikikomori tendency was mostly the same, except when questioning about the physical withdrawal by the frequency of going outside. The participant answered the robot with '1 day a week or less' and in the human interview her answer was '2-3 days a week'. In the survey, she responded: “The robot was easier to talk to than the human interviewer, with-out [sic] being nervous” [35].

Hereupon, one could theorize that hikikomori patients are more self-disclosing towards robots. This could tend to be an approach to counter the restraint of hikikomori or their parents due to social anxiety or shame [*Haji*] rooted in Japanese culture.

Lastly, there are suggestions that pet-like robots, which have been found to be effective in tackling social and mental impairments like dementia, anxiety and loneliness, could be applicable to hikikomori.

3.3 Potentially relevant robotics and technology

Although there is little research regarding hikikomori intervention, there is a substantial amount of work being done in Japan on using technology and domestic robotics for improving the well-being of the elderly, who, like hikikomori, have a tendency of suffering from loneliness or psychiatric disorders.

To begin with, there have been established good guidelines on how to design a robot for the user's need [3, 17]. These ensure that the needs and preferences of the elderly are at the forefront of their development process and play a big role in the user's acceptance and perceived enjoyment of the robot [8]. These user-centric approaches have created a wide variety of robots ranging from companion, conversational, humanoid and socially assistive robots. A comparatively smaller but nonetheless promising domain comprises chatbots and co-existing entertainment furniture. Chatbots offer social support, independent of time and place, either as appraisal, informational or emotional assistance. Especially with the good progress in artificial intelligence (AI), chatbots are capable of connecting through natural language user interfaces as well as performing fast emotion detection, allowing a personalized experience [1]. Adding vision using cameras, results in an even more personalized experience, due to the ability to gain social intelligence autonomously without relying on human input [24]. By processing this information, robots could distinguish if its user is in need of attention and is also willing to start an interaction to answer this need [23]. The most common therapeutic robots additionally include a selection of the following capabilities: 1) humanoid- or animal-like body 2) touch-, audial-, and depth-sensors 3) telepresence 4) text-to-speech and speech-to 5) virtual face display 6) ability to move independently [20]. If designed according to purpose, they can reduce loneliness [32] and many symptoms of dementia, such as anxiety, depression and nighttime behavior disturbance, while improving mood, physical activity and social engagement. [4, 10, 21]

Paro (Fig.2), a seal robot specifically designed for therapy and primarily used with dementia patients, has a positive effect on attitudes and emotions. Covered in soft fur and equipped with tactile sensors for touch, light, sound, and posture, *Paro* promotes interaction and has been shown to increase happiness, decrease depression and stress, and reduce anxiety in older adults, particularly those residing in care facilities. Its ability to simulate lifelike movements, such as neck and fin motion, contributes to a more engaging and therapeutic experience [19, 21]. However, its effect was short term (1-5 weeks) [4].

Amigus (Fig.3), a social robot designed to provide companionship and support to students, offers challenges, motivational recordings, and tasks to discourage isolation and promote mental well-being. By allowing users to express their emotions through buttons or gesture recognition, *Amigus* responds with empathetic eye expressions and plays corresponding audio clips from a predefined list. Additionally, the robot engages in conversations, encourages self-reflection, and stimulates communication, contributing to the overall psychological well-being of university students [3].

Tabby (Fig.4), a lamp prototype designed to provide coexisting entertainment content for everyday life, offering both direct and indirect interactions, aiming to maintain a seamless connection to

the user. Its lampshade, made from soft fur, creates a consciously breathing motion, accompanied by a pulsating light, giving the impression of it being alive. When touched, the light gradually darkens and returns to its normal state within approximately 3 seconds. Simultaneously, the lampshade fills with air from a full-speed fan, creating a bouncing sensation against the user's hand. Its breathing adjusts based on the frequency of touch and environmental noise. The filament lamp used in Tabby generates heat, creating a warm haptic impression [30].

Action Reproducer, (Fig.5) is a virtual reality system designed to address people's reduced walking, caused by a pandemic or social withdrawal. It provides users with a safe and immersive experience by combining a motion seat and slider-pedal devices to provide vestibular and lower-limb motion. A wearable pseudo force device and an avatar holding one's virtual hand create pulling sensations, allowing users to enjoy extended walking activities with reduced burden by simply sitting on the motion seat and perceiving multiple sensations [31].

In summary, there are though three important facts to consider, when trying to transfer these outcomes to hikikomori. First, most of the therapeutic research on the elderly was centered on 60–90-years-old people with dementia and in nursing homes or day care centers [23]. Second, the attempts to increase social engagement often referred to socialization within a care facility. Third, research often uses participants not from Japan or uses participants from Japan and other countries.

4 RESULTS

The research conducted on conventional intervention of hikikomori has revealed a preference of psychotherapy over pharmacotherapy [27], in-person over telepsychiatry treatment and mental health specialists over primary care providers [10]. Due to the importance of the robot design, such insights are important when designing therapeutic robots for hikikomori.

Such robots aren't yet capable of sustaining long-term treatment for mental health issues or hikikomori on their own. It is, however, unlikely that hikikomori sufferers will seek treatment through self-motivation [10]. Therefore, many guidelines recommend multiple steps to follow, with the main one being family intervention. In cases without family support, this position needs to be replaced [13, 16, 26].

Despite the limited research on hikikomori intervention, there exists a substantial amount of epidemiological and phenomenological research [34], which reveals that social anxiety and major depressive disorders are the most common disorders related to hikikomori [28]. A study in mental Japanese health and welfare centers has shown that one-third of hikikomori have autism disorder [15]. Pet robots, which are often used in research regarding their impact on elderly patients, have proven to have a positive impact on psychiatric disorders, especially autism and dementia [10]. These symptoms and disorders, despite neither being equal nor having the same origin, show a significant intersection, which indicates new directions of research.

5 DISCUSSION

One significant drawback is that the research on hikikomori focuses on young individuals and adolescents, whereas the research on the elderly focuses on individuals who are 60 years or older. This excludes those suffering and heading towards a lonely death [kodokushi], due to the 80-50 problem, which are between the two age groups being researched. With the incidence rate of hikikomori aged 40 to 64 being 1.45% - an estimated number of 613,000 people - this results in a big group of hikikomori not being included in research. Furthermore, the age at which people withdraw from society is evenly spread across all age groups without any major deviation [2].

Hikikomori's unwillingness to seek help as well as the severity of their conditions before dying in solitude, due to the (financial) inability to suddenly care for themselves, after their parents have passed away, surely play a crucial role regarding the amount of research done with this group.

Another difficulty is that, independently of the promising intervention offers, it can take months of family support and encouragement before hikikomori are willing to accept help [15]. Two-thirds of socially withdrawn youths do not consider meeting health professionals [6]. Even when they start, they often don't show up or stop going without notice [9].

Pokémon Go has proven to help hikikomori take this first step without external pressure. An interesting proposal is to create in-game points of interest, handing out rewards, at hikikomori support centers, which already exist in each prefecture in Japan but aren't compelling enough to bring all hikikomori out of their rooms [25].

However, conventional hikikomori therapy is time-consuming, little promising, and is carried by a shrinking labor force. Therapeutic, companion and conversational robots seem to be a viable alternative method, due to their ability to quickly adapt to their users' needs and take on a big workload.

Investing in such research now could be profitable, due to Japan's government orientating towards a futuristic society (Society 5.0) that emphasizes the integration of AI and robotics to create a human-centered society, where advanced technologies are harnessed to solve many problems, such as social challenges. It envisions a future where AI and robots play pivotal roles in transforming healthcare and improving overall well-being. They even have a group focusing on "[d]etection of early warning signs of [h]ikikomori" [7].

Future studies should examine the potential negative effects of technology and robotics on hikikomori - the goal is to help them interact and socialize with others outside their homes.

6 CONCLUSION

In conclusion, addressing the unique needs of hikikomori requires alternative approaches, and robotics offers a potential solution for their care. The prevalence of social anxiety, major depressive disorders, and autism among hikikomori emphasizes the need for tailored interventions. Pet robots, with their demonstrated positive impact on psychiatric disorders, present a promising avenue for hikikomori care. Further investment in research and development of therapeutic robots specific to hikikomori is essential.

SUPPLEMENTARY MATERIAL



Figure 2: Paro, the seal robot [17]



Figure 3: Amigus, a companion robot [3]



Figure 4: Tabby, a coexisting entertainment content [30]

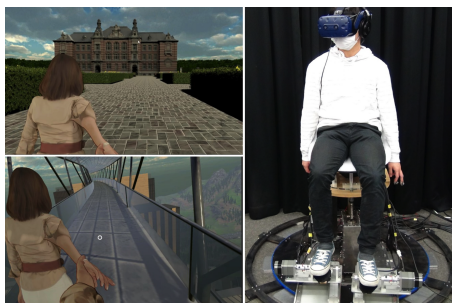


Figure 5: Action Reproducer, a virtual reality walking system [31]

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